

CMSC6950 — Fall 2025

Homework #2

Due Date: Work must be submitted to GradeScope or GitHub by 5pm (St. John's time) on Thursday, October 16.

- (5 points each) Perform each of the following tasks in a single Jupyter notebook. Submit a pdf export of the notebook to GradeScope for this problem.
 - Assign values to two variables that you cannot add. Show the error message from trying to add them.
 - Define a string variable for a sentence of at least 6 words. Use slicing to print only the third and fourth words. Use slicing to print only the last word.
 - We define `plex(n)` to be a 1 followed by `n` zeros, so that `plex(2)` is 100, and `plex(6)` is 1 million. Write a python function that takes an integer, `n` as input, with default value 0, prints a statement such as The plex of 3 is 1000. (correctly substituting the input value, `n`, and `plex(n)`) and returns the value of `plex(n)`. Run your code for 3 sample values of `n`, at least one of which is a single digit, and at least one which is greater than 20.
- (15 points) This exercise is intended to be completed in a single Bash terminal session. You should submit to GradeScope either one or more screenshot(s) of the session or copy/paste the terminal commands and output into a markdown cell in your Jupyter notebook from the first problem.

Perform the following actions:

- Make a directory named `Name_CMSC6950`, where `Name` is your name.
 - Change into the directory and run `git init`, checkout a new branch named `main`.
 - Edit a text file named `HW2.txt` and write three lines of text in the file (doesn't matter what).
 - Run `git status`, then add and commit `HW2.txt` to the repository and run `git status` again.
 - Edit `HW2.txt` to remove one line of text. Run `git diff` to show the changes. Add and commit `HW2.txt` to the repository.
 - Create a `.gitignore` file, to ignore all files with extensions `.dat` and `.log`, except for the file `important.dat`. Add and commit this to the repo.
 - Create three files to test the `.gitignore`, one of which is named `important.dat`, one with extension `.log` and one more with extension `.dat`. Try to add each of these in turn to the repository. Commit whichever file(s) you were successful in adding.
 - Show the one-line summaries of the git commits from the log.
- (Not to be handed in!) If you don't have one already, make an account on GitHub¹. A "free" account is sufficient for what you need for this course (and much more). If you haven't

¹<https://docs.github.com/en/get-started/signing-up-for-github/signing-up-for-a-new-github-account>

already done so, sign up for student benefits on GitHub². If you haven't already done so, generate ssh keys (on whatever system you're using for this class) and upload your public key to your GitHub account³.

4. (5 points) Go to <https://classroom.github.com/a/PipUiKog> and accept the "Intro to GitHub" assignment. When you click on this link, you should log in to github.com (if necessary) and then find and select your student number on the list of identifiers for the course. Note that there is nothing to hand in for this assignment - you will get these points for linking your GitHub account to your student number. Accepting this assignment will simply give you some additional information about GitHub and link your GitHub account to the GitHub classroom for this course.
5. (10 points) Go to <https://classroom.github.com/a/K-uMKIqb> and accept the "HW2 Python" assignment. If you've completed the above problem, you should be able to clone the repository, read the instructions in the README.md file, complete the assignment, and push your code to the repo to be graded. Nothing needs to be handed in to GradeScope for this problem.

²https://education.github.com/discount_requests/application

³<https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account>